

3(a)	protocist ;	1
3(b)(i)	any two from: 1 drugs have different, targets / mechanisms of action, or drugs, act on different, structures / processes / stages of life cycle ; 2 unlikely / less likely, for mutations against all drugs to occur ; 3 unlikely for resistance to occur against all the drugs ; I immune / immunity A described example 4 if there is resistance to one, the other drug(s) will, kill / AW, all pathogens ; AW I ref. to effective 5 (so) prevents, gene / allele, from being passed on ; 6 AVP ; e.g. low numbers left, so less probability of mutations arising <i>idea that</i> if all pathogens are killed, no replication / no mutations, for resistance to develop	2

Question	Answer	Marks
3(b)(ii)	<i>allow, pathogens / parasites, for <u>Plasmodium</u></i> I increased time when person is infectious <i>any two from:</i> increased time when person, has <i>Plasmodium</i> in, blood / circulation ; <i>allow mark if reference to, blood / circulation, not made but reference is made to blood in statements about Anopheles</i> A person acts as a reservoir for <i>Plasmodium</i> (so) quantity / concentration / number, of <i>Plasmodium</i>, remains high / increases ; AW A ref. to allows <i>Plasmodium</i>, to reproduce / replicate I ref. to genes passing to offspring increased, time / risk / chance / AW, of, being exposed to, vectors / mosquitoes / <i>Anopheles</i> (for blood meal) ; AW <i>idea of</i> more infected, <i>Anopheles</i> / mosquitoes available to feed on uninfected people ;	2

Question	Answer	Marks
3(c)(i)	1 correct use of examples ; <i>two of gene, gene mutation, protein, altered protein</i> <i>gene = kelch13</i> <i>gene mutation = kelch13 mutation</i> or F446I or (mutation has) adenine / A, not, thymine / T <i>protein/polypeptide = PfK13</i> <i>altered protein</i> isoleucine / ile, instead of, phenylalanine / phe 2 (mutation has), changed / altered / AW, sequence / arrangement / order, of nucleotides / bases, (in DNA) ; <i>ref. A replacing T still needs stated idea of a changed sequence</i> R if gene explanation also given and stated as sequence of amino acids 3 (gene) codes for the production of a, polypeptide / protein AW or (gene mutation) produces / results in, altered / changed / non-functioning / reduced function, polypeptide / protein ; AW A produces changed sequence of amino acids R if gene and gene mutation explanations given and one is incorrect	3

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3(c)(ii)	allow correct use of extracted data to help confirm points allow ideas such as higher survival = greater (partial artemisinin) resistance Key 20 / lower concentration = 20 nmol dm³ DHA 700 / higher concentration = 700 nmol dm³ DHA survival / survival rate = mean percentage survival rate of trophozoite any three from conclusions 1 to 5: <table><tr><td>1 zero survival of no mutation</td><td>cultures 1 <u>and</u> 4 / no mutations, no survival / 0% survival rate / all killed, at, higher concentration / 700 ;</td></tr><tr><td>2 concentrations compared</td><td>lower concentration / 20, higher survival rate than, higher concentration / 700 ora higher concentration / 700, lower survival rate than, lower concentration / 20 ; <i>general statement or can take one culture to compare</i></td></tr><tr><td>3 mutation (C580Y and F446I) v no mutation (control / cultures 1&4)</td><td>(A and B strains with) mutations <u>higher</u> survival rate than no mutation ora (A and B strains with) no mutation lower survival rate than mutations <u>;</u> <i>general statement for both strains, or A, or B</i> <i>(if concentrations mentioned, need to refer to both concentrations)</i></td></tr><tr><td>4 compare strain A with strain B</td><td>strain A higher survival rate than strain B ora strain B lower survival than strain A ; <i>general statement or comparing groups e.g. 1 and 4 or 2 and 5 or 3 and 6 (for both or one concentration)</i></td></tr><tr><td>5 C580Y v F4461</td><td>C580Y, higher survival than F446I / highest survival rate or F446I lower survival rate than C580Y ; <i>general statement or compare 2 and 3, or 5 and 6, for both or one concentration</i></td></tr></table>	1 zero survival of no mutation	cultures 1 <u>and</u> 4 / no mutations, no survival / 0% survival rate / all killed, at, higher concentration / 700 ;	2 concentrations compared	lower concentration / 20, higher survival rate than, higher concentration / 700 ora higher concentration / 700, lower survival rate than, lower concentration / 20 ; <i>general statement or can take one culture to compare</i>	3 mutation (C580Y and F446I) v no mutation (control / cultures 1&4)	(A and B strains with) mutations <u>higher</u> survival rate than no mutation ora (A and B strains with) no mutation lower survival rate than mutations <u>;</u> <i>general statement for both strains, or A, or B</i> <i>(if concentrations mentioned, need to refer to both concentrations)</i>	4 compare strain A with strain B	strain A higher survival rate than strain B ora strain B lower survival than strain A ; <i>general statement or comparing groups e.g. 1 and 4 or 2 and 5 or 3 and 6 (for both or one concentration)</i>	5 C580Y v F4461	C580Y, higher survival than F446I / highest survival rate or F446I lower survival rate than C580Y ; <i>general statement or compare 2 and 3, or 5 and 6, for both or one concentration</i>	3
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